

**SCHAEFER TECHNOLOGIES, INC.**

**OPERATION and  
SERVICE MANUAL  
for the**

**STI LABTOP BANDER**

**(CAPSULE BANDING MACHINE) Ver. 2.00**



**SCHAEFER  
TECHNOLOGIES, INC.  
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# STI LabTop Bander

## Capsule Banding Machine

### USER'S MANUAL

Valid from serial No. LB-17-179

## INTRODUCTION AND GENERAL RECOMMENDATIONS

### INTRODUCTION

This User's Manual has been compiled in compliance with Machinery Directive: 2006/42/EC and CE Marking Directive 93/68/EEC to provide the user with information about how to use and maintain the machine correctly. It is compulsory to read this manual. It is our intention that the information contained in this manual will be a valid aid for your work. This information is based on data and on our best current knowledge. Carefully read this manual, including the recommendations and suggestions. No part of this manual may be copied or passed to third parties without the prior written consent of Schaefer Technologies, Inc.

We advise you to carefully read this manual, as it is essential to install, check and service the machine correctly in order to keep it in a perfectly efficient working order. The manual contains tables, designs and diagrams, which will help you to get to know the machine in detail. Please don't hesitate to contact us if you find that an explanation has been omitted or is unclear.

### NAMEPLATE

For any servicing or overhaul operations not included in this manual or for any other technical problem which may arise, our Technical Personnel are available to provide you with information and to discuss any necessary action to take. When contacting us, please indicate the data on the machine's nameplate.

### PRELIMINARY REMARKS

- The drawings and designs which show the system should be considered as reference and not necessarily faithful in every detail.
- The system dimensions and specifications contained in this manual are not binding and may be changed without warning.
- The designs and all of the other documents supplied as part of the system are the property of Schaefer Technologies, Inc.

The system is covered by a guarantee and a contract of purchase. During the period in which the machine is under guarantee, any repairs which have not been authorized by Schaefer Technologies, Inc. and are carried out on the same machine will automatically make the warranty void.

The user must take particular note of the safety warnings and information indicated by the following symbols and labels in this manual.



Mandatory



Danger



Lockout



Prohibited



Read manual

**120 VOLTS**

Electrical Voltage Specification. 120 Volts



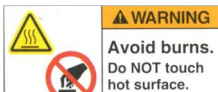
Electrical Current Specification. Single Phase

**220 VOLTS**

Electrical Voltage Specification. 220 Volts



Caution – Pinch Point. Moving parts below. Keep hands clear.



Warning – Avoid Burns. Do NOT touch hot surface. .



Warning – Hazardous Voltage. Risk of electrical shock or burn. Turn off and lock out system power before servicing.



devices or warning labels.

Warning – Avoid injury. Do not operate with guards removed. Replace guards before operating machine. For the safety of the Personal .DO NOT modify or remove any of this equipment's guards, safety



Warning – Read and understand operator's manual before using this machine. Failure to follow operating instructions could result in injury or damage to the equipment.

#### GENERAL SAFETY REGULATIONS

- THESE SAFETY REGULATIONS HAVE BEEN DRAWN UP FOR YOUR OWN SAFETY. The strict observance of the safety regulations will reduce the risks of accidents for both you and or others.
- DO NOT attempt to move, install or operate the system without having first read and fully understood this manual. If in doubt, ask your supervisor.

- Make sure that all of the guards and protective covers are in place BEFORE starting the machine.
- NEVER leave tools, mechanical parts or other unsuitable material on or in the machine.
- If a failure should occur, push the emergency stop button.  
NEVER WORK ON THE MACHINE WHILE IT IS RUNNING.
- Pay particular attention even when the main switch is in the “OFF” position, because the electrical power supply cables are still live.
- BEFORE restarting the production cycle after carrying out maintenance operations or repairs on the machine, make sure that all of the guards and protective covers are fitted correctly.
- Always be cautious, remember that your safety and that of fellow workers depend on you.
- When moving or lifting components of the machine, make sure that all the regulations for these operations are respected.

#### SPECIAL WARNINGS

- The personnel who operate this machine must know and scrupulously observe the general safety regulations. If these regulations are not observed, this may cause injury to workers and damage components of the machine.
- The maintenance operations must be carried out with the machine at a standstill. Lockout machine prior to servicing. Failure to follow operating instructions could result in personal injury or damage to the equipment.
- The user must make sure that all of the instructions in the manual are followed to the letter.
- Any tampering with the safety systems of the machine done for any reason, will be at the user’s own responsibility and risk.
- Do not ever try to bypass the safety systems or make them ineffective.
- For the machine smooth operation it is absolutely necessary to observe the speed, temperature and tension limits, along with all the indications given which the client must in any case respect.
- Furthermore, the environmental conditions must be taken into account for the installation of the machine.
- The national laws governing the use of this type of machine must also be observed.

## WARNING

Schaefer Technologies, Inc. declines any responsibility for damage which may derive from the non-compliance with the above-mentioned warnings.

The declaration of conformity enclosed with this manual has been compiled by the manufacturer in compliance with the European Directive and defines the machine as “safe”.

The manufacturer’s declaration will become null and void if the machine is tampered with, modified, has its fixed and moving protective guards removed or is used for purposes other than those for which it was designed. The manufacturer disclaims all responsibility for any accidents or damage that may arise.

### **SECTION I. GENERAL INFORMATION & SPECIFICATIONS**

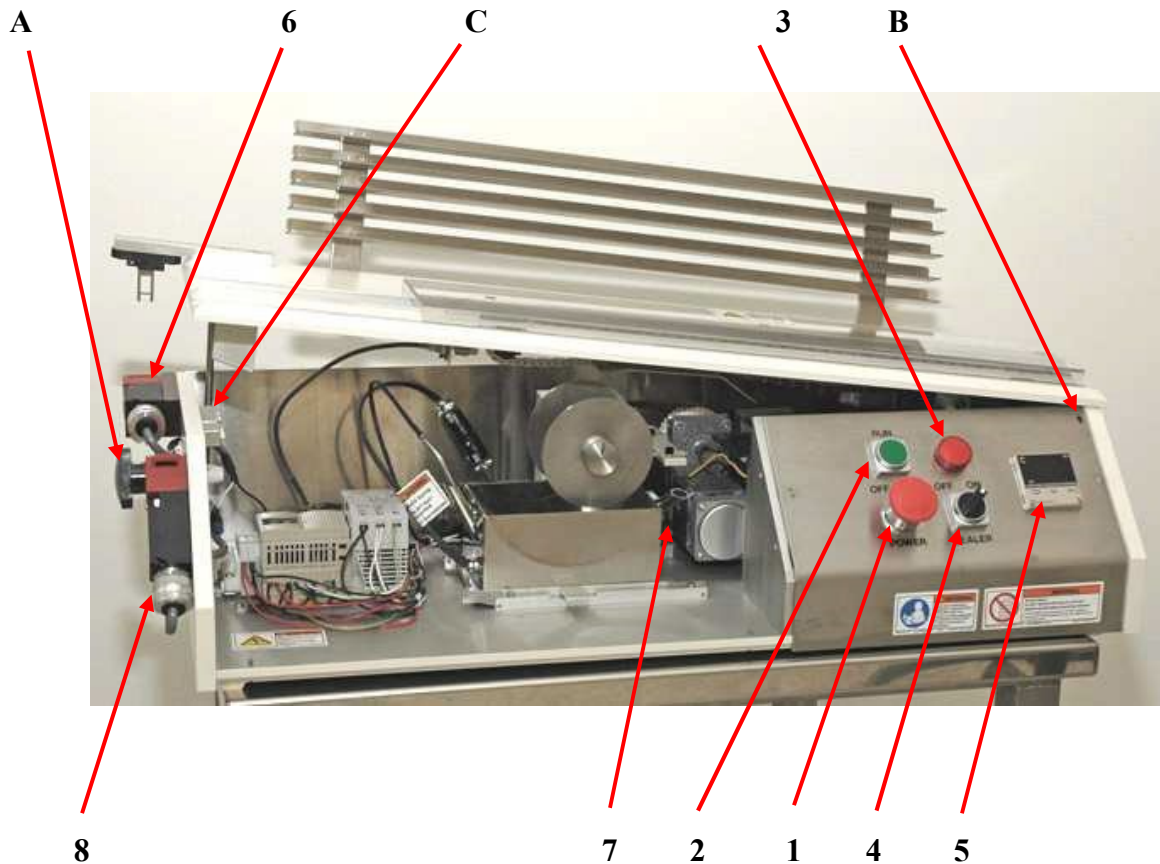
## INTRODUCTION



- **Note: Machine operation instructions are the same for the 120 volt and 220 volt machines in this manual. The only difference is the addition of a cooling fan and transformer for the 220 volt machine.**
- The Schaefer Technologies, Inc. LabTop Model Capsule Banding Machine is designed to band the joined portion of the cap and body of filled 2 piece hard capsules.
- Simulating the actual banding process of the production Capsule Banding Machine, it is now possible to make low volume runs for research and development or the evaluation of band colors.
- Capsule filled with oils, pastes, beads, or powders are loaded into the capsule carrying slats. The capsules are automatically positioned from the cap end, allowing the uniform banding of capsules with various closed lengths.
- The STI LabTop (Laboratory Model) Capsule Banding Machine uses a two-stage banding method. In the first stage, the capsules are positioned and banded around their circumference. When the first band is applied, occasionally air bubbles may be trapped in the gelatin and band may be uneven.
- After the second band is applied, the capsule carrier slat is manually transferred to the drying rack. The bands are dried at ambient conditions, minimizing moisture loss and affects on heat-sensitive ingredients.
- The final result is a uniformly banded capsule that is tamper resistant, and a stronger band than of one thick application. It is virtually impossible to open a capsule using this two-stage method without damaging or destroying the cap or body of the capsule.

## GENERAL INFORMATION &amp; SPECIFICATIONS

## COMPONENT LOCATIONS AND DESCRIPTIONS (120 volt version shown)

1. **ON-OFF (Push-off, turn clockwise-on) switch, main power**

This two-position switch controls all power to the unit.

2. **RUN (Push-button) switch**

Starts the capsule slat drive motor and hi-speed banding wheel drive motor.

3. **Pilot Light**

The pilot light will flash on and off to indicate that the operator is to transfer the slat from the front slat groove to the rear slat groove. When the pilot light comes on steady, the drive chain will traverse the tabletop. The pilot light flashes to indicate a low battery condition.

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GENERAL INFORMATION & SPECIFICATIONS**4. ON-OFF Switch, Banding Wheel**

The gelatin banding wheel switch is a two-position switch, which controls power to all motors and is to be turned on whenever gelatin is in the gelatin pan.

**COMPONENT LOCATIONS AND DESCRIPTIONS (CONTINUED)****5. Temperature Control and Indicator**

When the gelatin pan is properly in position, the temperature controller will display the heat transfer fluid temperature. The controller is front panel adjustable from 0 to 999°F.

**6. Interlock, Tabletop Safety**

If the table is not down and in its proper operating position, the interlock prevents the operation of the slat drive chain. NOTE: The banding wheels will continue to operate in low speed with this interlock disengaged. Use caution while performing work in this condition.

**7. Interlock, Gelatin Pan Heater Safety**

The interlock on the gelatin pan turns off power to the gelatin pan heating element and the low speed banding wheel motor if the gelatin pan is not raised into proper operating position.

**8. Interlock, Removable Panel Safety**

The interlock on the front panel stops the operation of the slat drive chain if the front panel is not in proper operating position. NOTE: The banding wheels will continue to operate in low speed with this interlock disengaged. Use caution while performing work in this condition.

**A. Vertical Adjustment Knob**

The tabletop has two points of adjustment. The first is vertical, which determines how much of the banding wheels will protrude through the tabletop. The amount of wheel protruding through the tabletop will determine the amount of banding solution that will be transferred to the capsule.

**B. Horizontal Adjusting Screw**

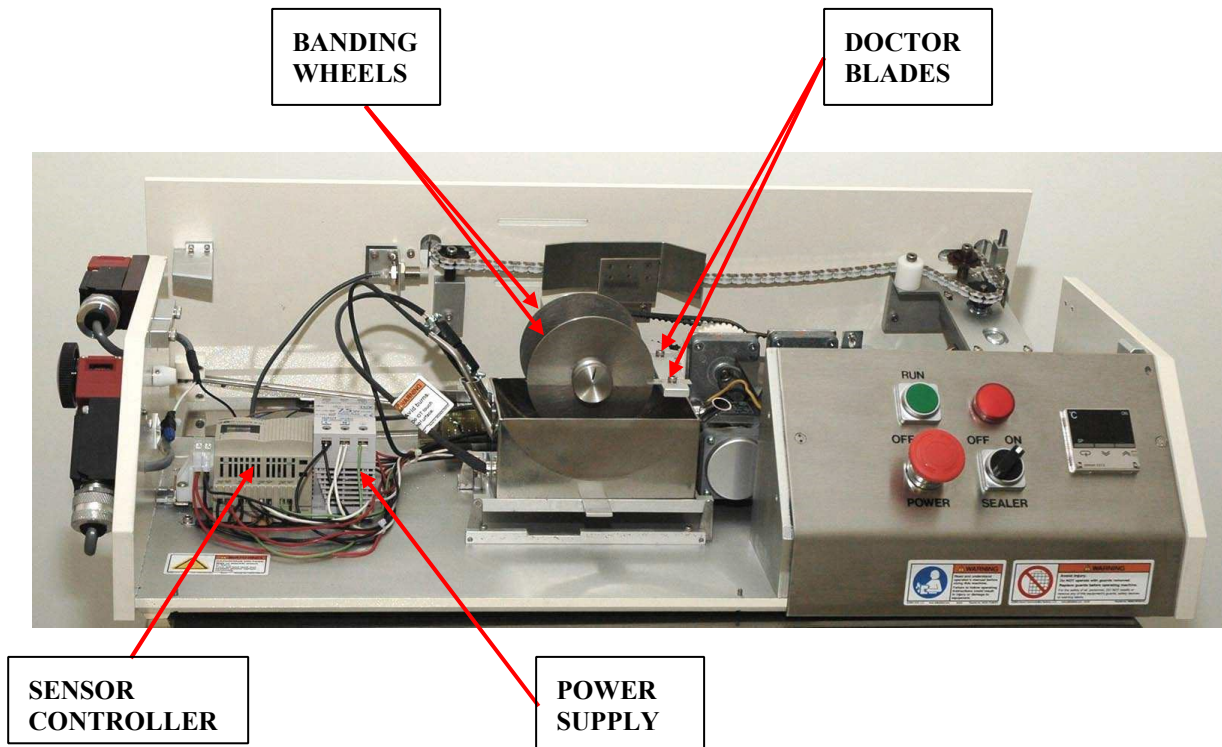
The second adjustment is horizontal and will determine the position of the banding band with respect to the cap/body joint line.

**C. Horizontal Adjustment Blocks**

The second adjustment is horizontal and will determine the position of the banding band with respect to the cap/body joint line.



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**GENERAL INFORMATION & SPECIFICATIONS****COMPONENT LOCATIONS AND DESCRIPTIONS (CONTINUED) (120 volt version shown)****BANDING WHEEL ASSEMBLY**

The banding wheels transfer gelatin from the gelatin pan to the circumference of the hard gelatin capsules.

- a) Wheels and spacers are a matched set as shown by their marking on the respective parts and must not be interchanged.
- b) Wheels and spacers are to be assembled and installed so that the identification markings on the banding wheels face the front of the machine.

**USE AND CARE OF COMPONENT PARTS**

The Schaefer Technologies, Inc. LabTop Capsule Banding Machine has been designed for low volume precision banding of hard gelatin capsules. It is very important that care be taken in handling and using this machine.

**Banding Wheel**

The banding solution transfer wheels (banding wheels) are precision parts and must be handled with extreme care. If the surface or edges on the circumference of the wheels become damaged, replace the wheel(s).

**Capsule Carrier Slat**

The capsule carrier slats are machined from solid aluminum.

To clean the carrier slats, a mild detergent and warm water should be used. Always store the carrier slats bottom down, on a flat, hard surface.

**Heating Element**

The heating element should be handled in such a way that minimal flexing occurs to the electrical cable.

NOTE: Discoloration is normal after prolonged use.

**Thermocouple**

The thermocouple should be handled with care in such a way that minimal flexing occurs to the cable.

**Sensor-Controller**

On models with replaceable batteries, replace lithium battery every five years. See Appendix A section for replacement of the battery.

## **ELECTRICAL PARTS LIST (continued)**

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### **SECTION I. GENERAL INFORMATION & SPECIFICATIONS**

#### **SPECIFICATIONS**

1. Capable of banding capsule sizes:

000,00, 0, 1, 2, 3, 4

2. Power Requirements:

120 VAC, 50/60 Hz, 3 Amps (US Model)

220 VAC, 50/60 Hz, 3 Amps (EU Model)

3. Recommended Ambient Room Conditions:

24° - 26°C (75° - 78°F) at 50-55% R.H.

4. Gelatin Characteristics:

Viscosity                      250-300 cps

Temperature Range    52°-55°C (125-130°F)

5. Overall Dimensions

Width 29" x Depth 12" x Height 12"

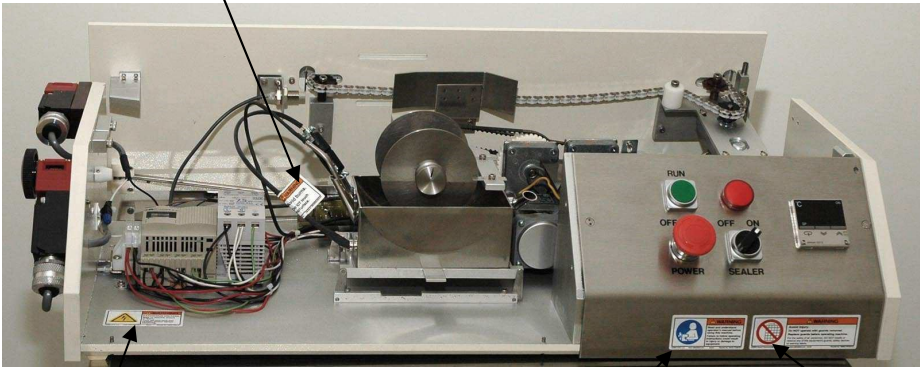
(736.6 mm x 304.8 mm x 304.8 mm)

6. Weight

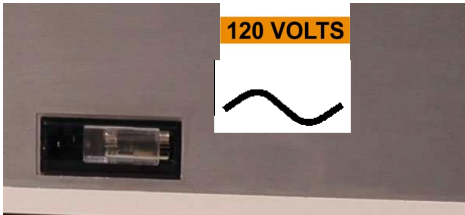
65 pounds (29.45 Kg)

ELECTRICAL PARTS LIST (continued)

Warning Labels and Locations



(120 Volt Version)



120 Volt Version



220 Volt Version  
OPERATING INSTRUCTIONS

SECTION II.

## ELECTRICAL PARTS LIST (continued)

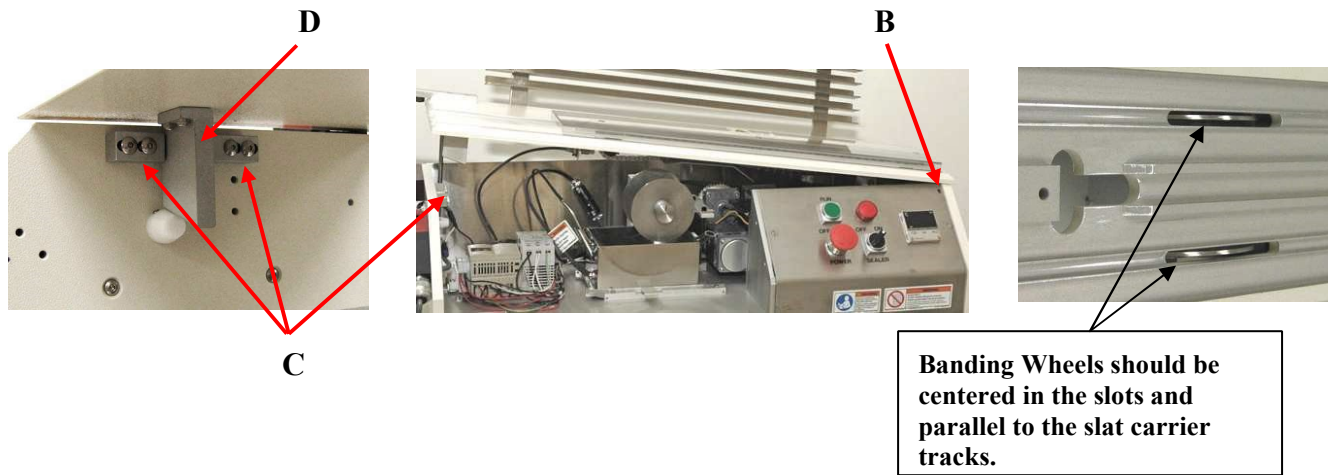
### BAND SIZE AND POSITIONING (TABLETOP ADJUSTMENTS)

#### Adjusting Band Size

The tabletop has two points of adjustment. The first is vertical, which determines how much of the banding wheels will protrude through the tabletop. The amount of wheel protruding through the tabletop will determine the amount of banding solution that will be transferred to the capsule.

#### Adjusting Band Position

The second adjustment is horizontal and will determine the position of the banding band with respect to the cap/body joint line. Adjustments necessary to reposition the banding band are as follows:



#### TOWARD CAP END:

1. Loose the screws holding the two slotted blocks, Item C.
2. Rotate front adjusting screw one quarter of a turn counter-clockwise, Item B).
3. Rotate the rear adjusting screw (accessible through the back panel directly behind front screw) one quarter of a turn clockwise.  
The adjustment described above changes the position of the banding band by 0.012 inches in the direction of the cap end of the capsule.
4. Visually align the table top so that the banding wheels and the table slots are parallel.
5. Slide the two blocks, “Item C” close but not tight against “Item D” and tighten the 4 screws.
6. Raise and lower the lid. If the lid binds adjust the “Item C” blocks to provide a little more clearance.

## ELECTRICAL PARTS LIST (continued)



### TOWARD BODY END:

1. Loose the screws holding the two slotted blocks, Item C.
2. Rotate the rear adjusting screw one quarter of a turn counter-clockwise.
3. Rotate front adjusting screw on quarter turn clockwise.  
The adjustments described above change the position of the banding band by 0.012 inches in the direction of the body end of the capsule.
4. Visually align the table top so that the banding wheels and the table slots are parallel.
5. Slide the two blocks, “Item C” close but not tight against “Item D” and tighten the 4 screws.
6. Raise and lower the lid. If the lid binds adjust the “Item C” blocks to provide a little more clearance.

## SECTION II.

## OPERATING INSTRUCTIONS

### BANDING PREPARATION

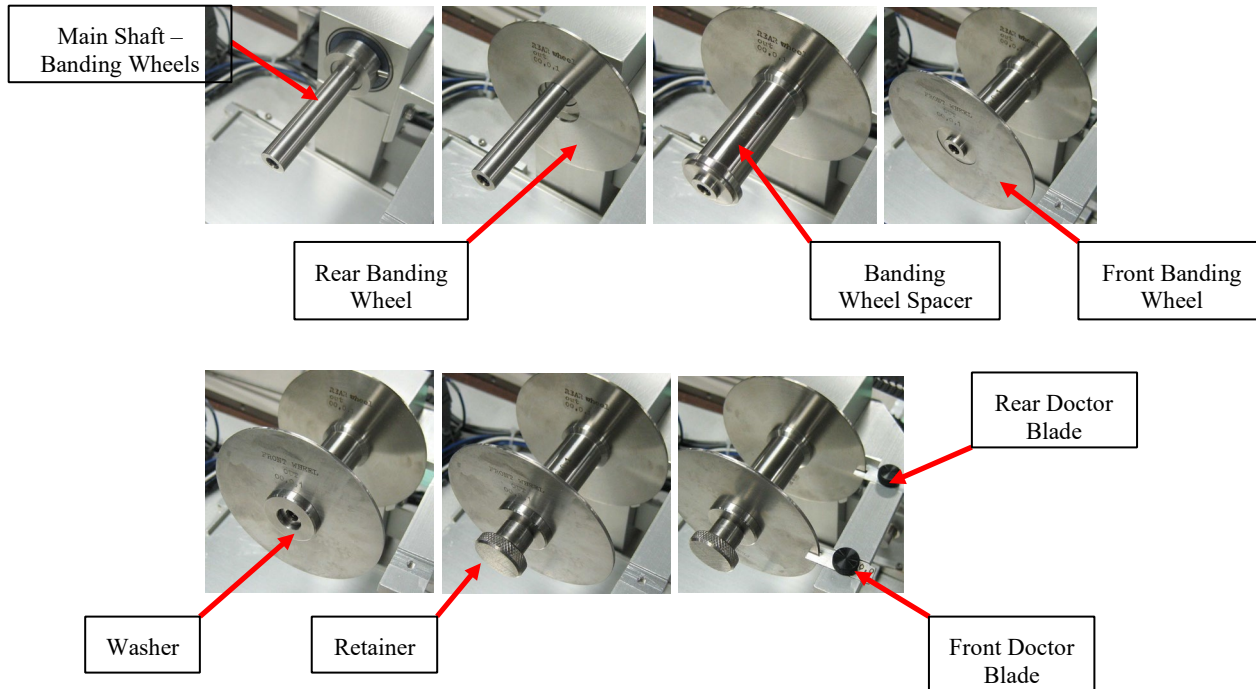
In preparing to band capsules, match the capsule carrier slat size to the actual capsule size.

The capsule carrier slat size appears on the top surface at the right end of each carrier slat, for example, “SIZE 00.”

1. Fill jacket of gelatin pan through straight end wall port with 300 ml of propylene glycol or distilled water, until visible in bottom of opposite port (90° elbow).
2. Insert heating element through tube in bottom of gelatin pan.
3. Position gelatin pan on pan base (with 90° port facing front of machine).
4. Assemble banding wheels (size markings facing out), spacer and doctor blade as indicated on wheels and spacer.



## ELECTRICAL PARTS LIST (continued)



### SECTION II.

### OPERATING INSTRUCTIONS

## **BANDING PREPARATION (CONTINUED)**

5. Place the rear banding wheel on the main shaft (size marking facing out).
6. Slide the banding wheel spacer on the main shaft until it engages in the through hole of the rear banding wheel.
7. Place the front banding wheel on the main shaft (size marking facing out) until it engages on the spacer.
8. Place the washer on the shaft and secure with the retainer screw.
9. Place doctor blades in locating grooves and secure in position. Allow approximately .010 spacing between the O.D. of wheel and bottom of slot in the blade. Also, there should be equal clearance between the slides of the slot and the side of the banding wheels.
10. Fill a slot with three capsules that are to be banded. Move the slot from left to right over the front banding wheel checking to make sure that the capsule cut line is centered on the banding wheel.
11. Transfer the slot to the rear groove; again check the alignment of the cut line to the wheel center.
12. With the slot filled with three capsules, place the slot in the front groove and slide the slot from the left-end towards the right-end, positioning a capsule over the “top dead center” of the front banding wheel. Raise or lower the tabletop by screwing the adjustment screw clockwise or

## **ELECTRICAL PARTS LIST (continued)**

counter-clockwise respectively until the capsule begins to rotate 100%. Rotate the adjustment screw one-half turn counter-clockwise. You are now ready to test banding capsules. It may be necessary to lower the tabletop to increase (or raise the tabletop to decrease) the amount of banding solution that is being transferred to the capsules.

13. Plug heating element into receptacle. Insert thermocouple into port in left end of gelatin pan.
14. Raise pan base to “full up” position.
15. Turn power switch (CW) to the “on” position.
16. Set temperature controller to 57°C (135°F)
17. Allow heat transfer fluid in gelatin pan to warm to preset temperature.

## **SECTION II. OPERATING INSTRUCTIONS**

### **BANDING PREPARATION (CONTINUED)**

18. Lower gelatin pan to the “down” position. (Banding wheels should stop rotating).
19. Fill gelatin pan to indicator line on back wall in tub area.
20. Place gelatin pan cover on top of pan (must be used at all times).
21. Raise gelatin pan base back into the “full up” position. (Banding wheels should begin to rotate at low speed).
22. Allow five (5) minutes for banding wheels to warm.  
Note: To add gelatin during operation, reverse step 21 and 20, repeat step 19  
PLEASE NOTE: When running the heating cartridge for the first time, it is common to see a light smoke emitted from the heating element. This should subside after a short time. If not, contact STI. Discoloration of the heating element is common after use.



## ELECTRICAL PARTS LIST (continued)

### BANDING OPERATION

1. Properly position a capsule carrier slat filled with capsules in the front slat groove at the left-end (slat end to be flush with end of tabletop).
2. Press the “Run” push button. The pilot light will illuminate. The slat chain drive and the high speed banding wheel motors are activated. As the slat passes over the front banding wheel the first band is applied.
3. The pilot light turns off and the slat chain drive and high speed banding wheel motors stop after the banding pass. The operator then transfers the slat to the left-end of the rear slat groove for the second band application. Note: At this time a new slat can now be positioned in the left-end of the front slat groove.  
Pressing the “Run” push button starts the banding pass.
4. When the pilot light goes out, the slat chain drive and high speed banding wheel motors have stopped, the operator moves the rear position slat to the drying rack for curing and the loading process can be repeated.

## SECTION II.

## OPERATING INSTRUCTIONS

### CHANGING CAPSULE SIZES

1. The capsule size number on the slat must be the same as the capsule size.
2. Banding wheel and spacer identification is as follows:

#### Size 000,00, 0, & 1

| <u>Part</u>   | <u>Marking</u> |
|---------------|----------------|
| Rear Wheel    | 00-0-1         |
| Front Wheel   | 00-0-1         |
| Spacer        | 00-0-1         |
| Doctor Blades | 00-0-1         |

#### Size 2, 3, & 4

| <u>Part</u>   | <u>Marking</u> |
|---------------|----------------|
| Rear Wheel    | 2-3-4          |
| Front Wheel   | 2-3-4          |
| Spacer        | 2-3-4          |
| Doctor Blades | 2-3-4          |

The banding wheels are to be mounted so that the identification markings face the front of the unit.

**SECTION II.**

**OPERATING INSTRUCTIONS**

**CLEAN UP PROCEDURE**

1. Turn power off, remove front left panel and raise tabletop.
2. Lower gelatin base pan.
3. Remove gelatin pan cover.
4. Unplug heating element.
5. Remove thermocouple.
6. Remove both doctor blades.
7. Remove banding wheel assembly.
8. Insert rubber plugs in both ports in end wall of gelatin pan.
9. Remove gelatin pan from the machine.
10. Remove heating element from gelatin pan.
11. Remove unused banding solution.

Wash all contact parts with HOT water. Be sure the doctor blade holder is cleaned thoroughly.

**SECTION III.**

**SERVICE PROCEDURES**

**TROUBLESHOOTING**

1. Off-set banding band
  - a. Wrong slat size or wrong banding wheel set.
  - b. Incorrect horizontal adjustment of tabletop, see Section II, band Size and Positioning.
2. Big banding band, bubbles
  - a. Tabletop too low. See Section II, Band Size and Positioning.
  - b. Heavy viscosity.
  - c. Foaming of gelatin.
3. Small or partial band

## **ELECTRICAL PARTS LIST (continued)**

- a. Tabletop too high. See Section II, Band Size and Positioning.
4. Capsules do not rotate on friction rails
  - a. Wrong slat size.
  - b. Be sure pockets in slats are clean.
  - c. Be sure friction rails (2) are clean.
5. Machine will not operate, pilot light not on when run button is pressed.
  - a. Check the front cover tabletop lid for proper positioning.
  - b. Check the power cord.
  - c. Check condition of the main fuse located in the power entry module.
  - d. Check power switch, turn clockwise and release to “On” position.
  - e. Raise tabletop lid and check “Run/Hold” LED on sensor controller. If the LED is off or flashing, service should be referred to Schaefer Technologies, Inc.
  - f. If “Run/Hold” LED on sensor controller is on, check for loose connections. If connections are not loose, service should be referred to Schaefer Technologies, Inc.

## **SECTION III. SERVICE PROCEDURES**

### **TROUBLESHOOTING (CONTINUED)**

6. Banding wheels not turning when gelatin pan is installed and raised to “Up” position.
  - a. Dried gelatin on doctor blade, see Section II, Clean Up Procedure.
  - b. Bander switch must be in the “On” position to run gelatin wheels.
  - c. If Bander switch is “On,” check input on sensor controller. If not “On,” check connections, adjust, replace sensor as required.
  - d. If input LED is “On,” check output LED. If LED is not “On,” contact Schaefer Technologies, Inc.
  - e. If output LED on is “On,” check Opto22 module #3 on the Opto22 mounting rack. If the module’s LED is “On,” check pico fuse adjacent to Opto22 output module. Replace as required.
7. No display on temperature controller when gelatin pan is installed and raised to “Up” position.
  - a. Check gelatin pan sensor. See Section II, Troubleshooting.
  - b. Check Opto22 module #4. If LED is “On,” check, replace pico fuse or Opto module as needed.
  - c. If Opto22 module and fuse are ok, check thermocouple, replace as required.
8. Temperature controller display does not control at preset temperature.

## ELECTRICAL PARTS LIST (continued)

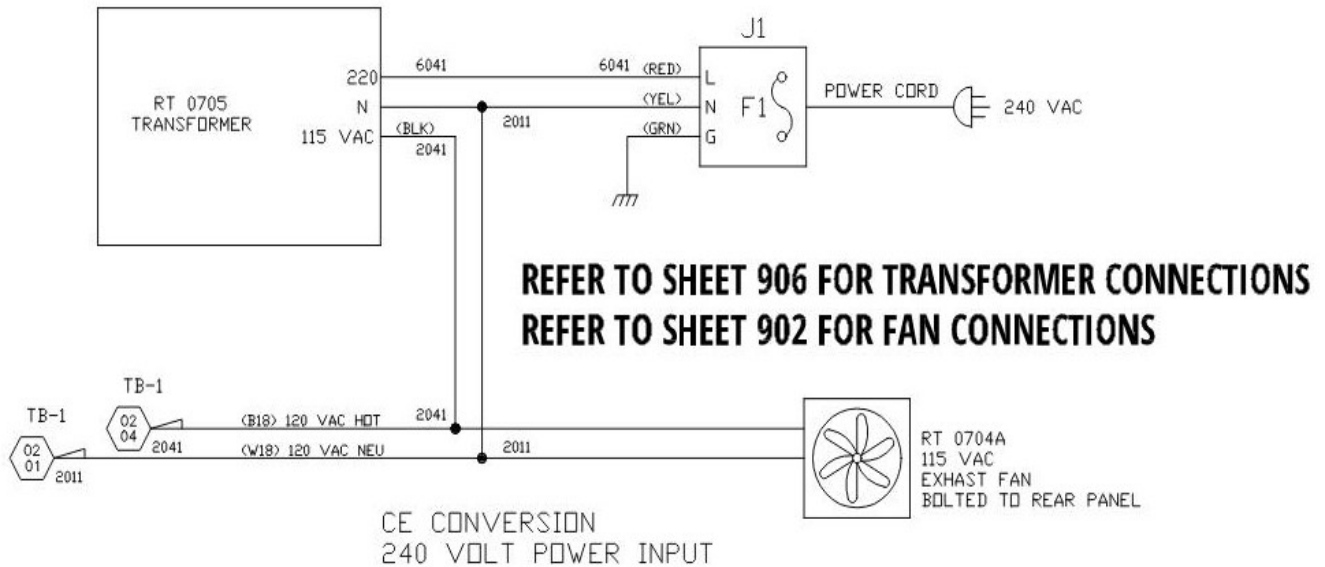
- Allow temperature to equilibrate.
- If the temperature on the digital display is above the set point, turn the reset adjustment clockwise.
- If the temperature on the digital display is below the set point, turn the reset adjustment counter-clockwise.

**Make small adjustments and allow the temperature to equilibrate between each adjustment.**

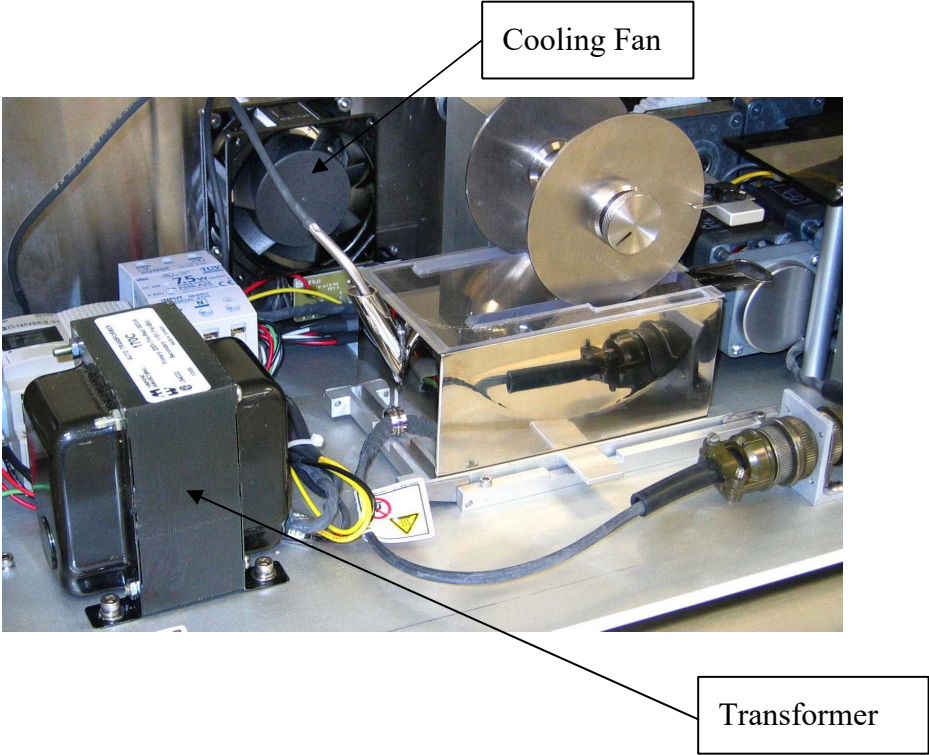
### SECTION III.

### SERVICE PROCEDURES

#### **EU POWER CONVERSION (220 Volt Modification)**



**ELECTRICAL PARTS LIST (continued)**



**SECTION III. SERVICE PROCEDURES**

**ELECTRICAL PARTS LIST**

| ITEM | REQUIRED | PART NUMBER | DESCRIPTION                               |
|------|----------|-------------|-------------------------------------------|
|      | 1        | RT0502      | Temperature Controller                    |
|      | 1        | RT0503      | Temp. Controller Socket                   |
|      | 1        | RT0504      | Thermocouple                              |
|      | 1        | RT0537      | Cartridge Heater and Heater Plug Assembly |
|      | 2        | RT0508      | Chain Sensor                              |
|      | 1        | RT0512      | I/O Module Mounting Rack                  |
|      | 4        | RT0513      | AC Output Module                          |
|      | 1        | RT0515      | Pilot Lamp 120V,1.8W                      |
|      | 1        | RT0516      | Pilot Light Lens                          |
|      | 1        | RT0517      | Light Module                              |
|      | 1        | RT0518      | Power Switch Actuator                     |
|      | 1        | RT0519      | Contact Block                             |

**ELECTRICAL PARTS LIST (continued)**

|  |     |          |                              |
|--|-----|----------|------------------------------|
|  | 1   | RT0520   | Banding Switch Actuator      |
|  | 1   | RT0522   | Start Button Actuator        |
|  | 3   | RT0528   | Gear Motor                   |
|  | 3   | RT0529   | Oil Filled Capacitor         |
|  | 3   | RT0530   | Capacitor Bracket            |
|  | 1   | RT0533   | Fuse 3 amp volt              |
|  | 1   | RT0700   | Plug for Heater              |
|  | 1   | RT0701   | Receptacle for Heater        |
|  | 2   | RT0702   | Cable Cover for Heater       |
|  | 2   | RT0703   | Cable Grip for Heater        |
|  | 3ft | CF2817   | 8770 Cable                   |
|  | 2   | CF3095   | Ground Terminals             |
|  | 1   | AU1979   | Fuse 1-Pole                  |
|  | 1   | AU1980   | Fuse Holder                  |
|  | 1   | RT0688   | Power Entry Mounting Bracket |
|  | 1   | RT4958   | Safety Cover                 |
|  | 2   | RT0670A  | Clutch Bearing               |
|  | 2   | RT0670B  | Gear Belt Pulley             |
|  | 2   | CF2840   | Key Switch                   |
|  | 2   | CF2841   | Key                          |
|  | 1   | CF2847   | Switch Spacer                |
|  | 2   | CF2750   | Hubbel                       |
|  | 1   | RT0700-1 | Heater-Plug                  |
|  | 1   | RT0701-1 | Heater-Socket                |
|  | 2   | RT0702-2 | Cover Plug & Socket          |
|  | 2   | RT0703-2 | Clamp                        |

**ELECTRICAL PARTS LIST (continued)**

|  |   |        |                   |
|--|---|--------|-------------------|
|  | 1 | RT0686 | Key Bracket Top   |
|  | 1 | RT0687 | Key Bracket Front |
|  | 1 | RT0697 | Manual            |

**MECHANICAL PARTS**

| ITEM | REQUIRED | PART NUMBER | DESCRIPTION              |
|------|----------|-------------|--------------------------|
|      | 1        | RT0600      | Base Plate               |
|      | 1        | RT0602      | Gelatin Pan Sensor Mount |
|      | 1        | RT0604      | Top Plate                |
|      | 1        | RT0606      | Guide                    |
|      | 1        | RT0607      | Mount, Return Idler      |
|      | 1        | RT0608      | Spacer                   |
|      | 1        | RT0609      | Washer                   |
|      | 1        | RT0610      | Idler, Return            |
|      | 1        | RT0611      | Idler                    |
|      | 1        | RT0612      | Drive Socket             |
|      | 1        | RT0613      | Left End                 |
|      | 2        | RT0614      | Link Front Weldment      |
|      | 2        | RT0615      | Link Back Weldment       |
|      | 1        | RT0616      | Spacer 00, 0, 1          |
|      | 1        | RT0617      | Gelatin Tub Base         |
|      | 1        | RT0618      | Bearing Mount            |
|      | 2        | RT0619      | Switch Panel Mount       |

**MECHANICAL PARTS**

|                     |   |        |                                 |
|---------------------|---|--------|---------------------------------|
|                     | 1 | RT0620 | Main Shaft - Gelatin Wheel      |
|                     | 1 | RT0621 | Mount, Front Link               |
|                     | 2 | RT0622 | Mount, Rear Link                |
|                     | 1 | RT0623 | Control Panel                   |
|                     | 1 | RT0624 | Front Panel, Removable          |
|                     | 1 | RT0625 | Drying Rack                     |
| EU Power Conversion | 1 | RT0705 | Transformer (used for 220 volt) |
| EU Power Conversion | 1 | RT0704 | 220volt exhaust fan             |
|                     | 1 | RT0629 | Rear Panel                      |
|                     | 1 | RT0630 | Right End                       |
|                     | 1 | RT0631 | Chain Cover                     |

|  |   |        |                      |
|--|---|--------|----------------------|
|  | 1 | RT0633 | Guide                |
|  | 1 | RT0634 | Motor Mount, Chain   |
|  | 1 | RT0635 | Weldment, Kick Stick |
|  | 1 | RT0636 | Gusset, Center       |
|  | 1 | RT0637 | Base, Motor Mount    |
|  | 1 | RT0638 | Pivot                |
|  | 3 | RT0639 | Thrust Washer        |
|  | 1 | RT0640 | Handle               |
|  | 2 | RT0641 | Hinge                |
|  | 1 | RT0644 | Holder, Kick Stick   |
|  | 1 | RT0645 | Guide                |
|  | 1 | RT0646 | Chain Side Weldment  |
|  | 1 | RT0648 | Idler                |



**SECTION III.****SERVICE PROCEDURES**

|  |   |        |                        |
|--|---|--------|------------------------|
|  | 2 | RT0649 | Stand-off              |
|  | 2 | RT0651 | Motor Mount            |
|  | 1 | RT0652 | Doctor Blade Holder    |
|  | 1 | RT0653 | Mount, Chain Prox.     |
|  | 1 | RT0654 | Chain Assembly         |
|  | 1 | RT0656 | Gelatin Pan Weldment   |
|  | 1 | RT0658 | Guide                  |
|  | 1 | RT0659 | Cover, Gelatin Pan     |
|  | 1 | RT0660 | Tip                    |
|  | 2 | RT0661 | Doctor Blade, 00-0-1   |
|  | 2 | RT0662 | Doctor Blade, 2-3-4    |
|  | 1 | RT4990 | Slat, Size 0 (15 Row)  |
|  | 1 | RT4992 | Slat, Size 00 (15 Row) |
|  | 1 | RT4995 | Slat, Size 1 (15 Row)  |
|  | 1 | RT4997 | Slat, Size 2 (15 Row)  |
|  | 1 | RT4999 | Slat, Size 3 (15 Row)  |
|  | 1 | RT5000 | Slat, Size 4 (15 Row)  |
|  | 1 | RT0683 | Spacer 2, 3, 4         |
|  | 1 | RT0671 | Gear Belt Pulley       |
|  | 1 | RT0672 | Gear Belt Pulley       |
|  | 1 | RT0673 | Gear Belt              |
|  | 1 | RT0674 | Gear Belt              |
|  | 2 | RT0675 | Retaining Ring         |
|  | 2 | RT0676 | Ball Bearing           |

**MECHANICAL PARTS (continued)**

| ITEM | REQUIRED | PART NUMBER | DESCRIPTION                  |
|------|----------|-------------|------------------------------|
|      | 3        | RT0678      | Shoulder Bolt                |
|      | 1        | RT0679      | Clamp Collar                 |
|      | 2        | RT0680      | Counter Sink Oval Head Screw |
|      | 2        | RT0681      | Set Screw-Nylok              |
|      | 4        | RT0682      | Bump-on                      |
|      | 1        | RT1983      | Gelatin Wheel Front 00, 0, 1 |
|      | 1        | RT1985      | Gelatin Wheel Rear 00, 0, 1  |
|      | 1        | RT1982      | Gelatin Wheel Front 2,3,4    |
|      | 1        | RT1984      | Gelatin Wheel Rear 2,3,4     |
|      | 1        | RT0684      | Knob                         |
|      | 1        | LF8027      | Din Rail                     |
|      | 1        | AU1020      | Power Cord                   |
|      | 1        | RT4961      | Zen controller               |
|      | 1        | RT4959      | 24V power supply             |
|      | 3        | LF8030      | End Barrier                  |
|      | 13       | LF8026      | Terminals                    |
|      | 2        | LF8028      | End Caps                     |
|      | 2        | RT4962      | Handle                       |